

Seeing the Forest for the Trees: Why Complex Interdependencies Need Not Inhibit Effective Search

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Abstract

The simulation-based literature on search and learning suggests that humans are unlikely to find superior solutions when the task environment has many local optima. Here, we show that this conclusion is an artifact of the widely-used *NK* performance landscape model. The eponymous *K* parameter that tunes the count of local optima, or *ruggedness*, alters several features of the landscape that can determine the outcome of the search process. After documenting those features, we propose the *Dirichlet Dot Product* model as an environment that can manipulate ruggedness without altering the other features. In a preregistered experiment, we find that participants' search performance on highly rugged Dirichlet Dot Product landscapes is as good or better as on single-peaked *NK* landscapes. Effective search can be compatible with rugged landscapes; when there are broader gradients present in the environment, human actors are not hopelessly myopic and can do well navigating around local peaks. This finding suggests a need to be cautious with the putative cognitive implications of the *NK* model and a need for more research on the macroscopic structure of common task environments.

Introduction

In a canonical statement in the study of human reason, “A Behavioral Model of Rational Choice,” Herbert Simon says our goal should be to “replace the global rationality of economic

man with the kind of rational behavior that is compatible with the access to information and the computational capacities that are actually possessed by organisms, including man, *in the kinds of environments in which such organisms exist*” (? , p. 99, emphasis added). The features of environments are key to understanding what information is available, yet research since then has largely focused on human computational capacities under strict assumption about access to information. Assessing the relationship between environments and information is necessary to gain a fuller understanding of human cognitive capacities.

To do that, Simon argued that a standard and well-understood task environment is a necessary research tool. A fan of the game of chess, Simon proposed it become the “*drosophila*” of psychology, the “standard task environment around which knowledge and understanding can cumulate” (Simon and Chase, 1973, p. 393). Yet, Simon and his coauthors could not account for certain “recalcitrant facts” about behavior in this environment. With the benefit of hindsight, we can see that chess could never be the *drosophila* of cognitive science. It engages only elements of our cognitive capacities; the great successes of computers in the realm of chess contrast starkly with the limited success in other domains, particularly those related to mobility (Mitchell, 2021). This highlights how diverse human capacities are, but also that we operate, cognitively speaking, in many different kinds of environments.

The social sciences need a variety of standardized task environments to better understand behavior, and one very promising environment that has received a lot of attention is the *NK* fitness landscape Kauffman and Weinberger (1989). Levinthal’s seminal paper (1997) introduced the notion of a fitness, or performance, landscape to the strategy and decision-making literature. By 2019, a review Baumann et al. (2019) identified 70 simulation-based articles in top journals using some version of the *NK* model. The model has proven useful because it parameterizes the degree of interdependencies in an array of choices that actors must make. Actors, particularly organizations, often face complex task environments in which there are many decisions to be made, and each choice can affect the performance contributions of other choices (Simon, 1962). For example, a laptop can be thought of as a combination of a screen, a battery, and a CPU. The value of the whole device depends on the interdependencies between components, because, for example, a large screen and a fast CPU do not amount to much if the battery

cannot support their use for long (Csaszar, 2018). The model elegantly links interdependent choices to performance values and showed that interdependencies tended to create “rugged” landscapes. A visual metaphor, ruggedness refers to a performance terrain with many peaks, or local optima. Theory and simulation-based evidence suggest that ruggedness renders boundedly rational actors ineffective in searching complex landscapes.

However, there has not been a systematic study of the NK landscape itself in this literature. The model establishes that interdependencies can generate the *microscopic* feature of ruggedness, but interdependencies are likely to alter other features of the environment that can affect search success. There has been no systematic assessment of how increasing interdependencies alters other features. Some have considered the nature of the underlying pattern of interdependencies (Rivkin and Siggelkow, 2007; Marengo, 2015), but not the diverse *macroscopic* features generated by the model. Interdependencies are the genotype to the model’s fitness phenotype, and knowing the former does not by itself reveal much about the latter. In one way, the lack of insight here is not surprising; visualizing the landscape is not possible due to the high number of dimensions typically used. Without such diagnostic aids, it is hard to develop intuitions about the macroscopic features. However, doing so is necessary because macroscopic features are likely to alter the effectiveness of search and the relative benefits of exploration and exploitation.

In this paper, we undertake two analyses. First, we show that K —the primary control parameter of the NK model—tunes several macroscopic features of the landscape in ways that impact the effectiveness of local search. Together, these changes mean that increasing ruggedness by increasing K is associated with a breakdown of any meaningful relationship between pairs of choices and the resulting performance, a “house-of-cards” model Kingman (1978). This is a plausible state of affairs for the protein folding problem the NK model originally dealt with, but it is implausible for most problems humans face. It also means that the poor search outcomes currently associated with ruggedness might have more to do with the other changes than ruggedness.

With that in mind, we introduce the Dirichlet Dot Product (DDP) model as a landscape that can vary ruggedness without significantly altering other features. It has a straightforward

implementation and can create landscapes with degrees of ruggedness comparable to the *NK* landscape. However, it maintains a broad gradient across the landscape. There is no such gradient in rugged *NK* landscapes, which means there is no signal for actors to attend to, just noise. Work in psychology shows that our attentional systems often act as gain control systems that reduce noise and amplify signal (Pashler et al., 2001). In landscapes, a gradient is the salience signal and local peaks the noise. Can human subjects see the “forest” of the gradient despite the “trees” of the local peaks?

We test whether this is the case in a preregistered human-subject experiment using *NK* and *DDP* landscapes of equivalent ruggedness. We replicate prior experimental results on *NK* landscapes and then find that participants can search highly-rugged *DDP* landscapes as effectively as they can search *unrugged, single-peaked NK* landscapes; despite the presence of numerous local peaks, participants search broadly and manage to approach global optimum. We conclude that ruggedness alone is not an impediment to effective search, and the results in the *NK* literature are driven primarily by the macroscopic features of that environment.

These findings indicate a need to expand the space of landscapes considered in the literature. Making inferences about cognitive and computational capacities without controlling for many emergent features of the landscape (instead of just a given control parameter) runs the risk of spurious conclusions about those capacities. The Dirichlet Dot Product landscape is only one alternative that captures some, but not all, relevant task environments. The undertaking of this work of expanding the space of landscapes will be vital in the advancements in how we think about human processes of search and learning.

The *NK* Performance Landscape Model

The theoretical study of performance in management and strategy science owes much to the study of fitness landscapes in evolutionary biology. Starting with Levinthal’s seminal study of organizational speciation (Levinthal, 1997), Kauffman’s *NK* fitness landscape became a workhorse model for the study of organizational performance and learning. Some environments for action do not exhibit interdependencies between choices and therefore do not present complex

problems to managers. For example, increasing the advertising budget for a local builder could increase the number of customers and net profits without altering labor costs, the quality of final product or reputation. Yet many firms and managers often have to contend with complex decision making scenarios where one decision affects the outcome of many others. How much of the total budget should be directed to marketing for a particular product? R&D on another product? Where should the next corporate office be located? These strategic or policy decisions can effect the firms overall performance through a web of interdependencies that decision-makers may not even be aware of.

Some choices may have a synergistic nature, as when the choice of office location makes it easier (harder) to hire engineering talent and improves (harms) R&D productivity. Others can offset each other. Directing more of the marketing budget to one particular product can improve its sales, but cannibalize those of another product. Shrewd decision-makers should be aware of some of these interdependencies and take them into account before making decisions, but when the number of interdependencies are high, such *apriori* reasoning becomes untenable. As the interdependencies increases, there are necessarily more negative complementarities between the individual decisions and this creates local optima, any set of decisions such that changing any one individual decision within the set lowers the overall performance. Following Kauffman, the management literature refers to the existence of many local optima as a “rugged” landscape.

As a parameterized model, the *NK* model generates families of landscapes with different counts of local optima (i.e., “ruggedness”) and thereby facilitates theorizing about the role of the degree of interdependencies and problem complexity. Motivating much of this research is the examination of optimal strategies with regards to exploration vs. exploitation (March, 1991). Other research has used *NK* or similar models to study search strategies of firms, how they adapt, and implications for population dynamics with regards to variation and selection firms in an industry (Levinthal, 1997; Levinthal and Posen, 2007; McKelvey, 1999; Rivkin, 2000; Lenox et al., 2006, 2007). Insights have been generated such as how variation in industries may emerge. Levinthal (1997) identified that in complex environments a multitude of heterogeneous firms may exist. Rivkin (2000), for example, identified how even when decisions may be publicly observed how imperfect imitation prevents effective competition. However, successful firms

and decision makers may struggle to adapt in complex landscapes and may even be selected out for more adaptable firms that may make inefficient decisions (Levinthal 1997; Levinthal & Posen 2007). Though heuristic strategies have been developed, such as how managers may create cognitive representations of landscapes which serve to reduce dimensionality of complex landscapes and aid in search (Csaszar and Levinthal 2016). A rich literature has been developed based on the *NK* model.

However insights from past research are dependent upon the particular type of interdependent complexity defined by and arising from the *NK* landscape. As Kauffman knew when he proposed the model, it describes an extremely hard problem to solve. But as an evolutionary biologist, he was only interesting showing that the exploration mechanism that is random genetic mutations is likely to do poorly in rugged fitness landscapes. Random mutations generally alter a single gene and evolutionary theory predicts that any mutation that moves an organism off a local fitness optimum will be selected out of existence. Without any neutral or positive adaptations possible, evolutionary progress is slow.

But humans do not only make individual random mutations or changes to their decision set so the question arises whether rugged landscapes are truly a challenge for human decision-makers to search. We can make big and knowledge guided jumps in the search space. We can learn the features of the problem space as we explore it. And yet simulation and human-subject studies suggest we also struggle to successfully search rugged *NK* landscapes and so the claim is that ruggedness presents a problem for human decision making.

Here we question that reasoning because it is clear that tuning “ruggedness” in the *NK* model also alters other features of the landscape that have plausible connections to search outcomes. To make that argument, we define the model and then look at how four particular features change as interdependencies are increased.

The *NK* model and Its Macroscopic Features

The *NK* model is a landscape of N binary decisions mapping onto fitness values for each decision. The K parameter in the *NK* model tunes the level of interdependency in the model; $K=0$

implies no interdependencies, $K=1$ signifies that every decision's fitness is dependent on one other decision.

The NK model is a useful model to start to better understand the relationship between performances outcomes and the number of interdependencies within the decision-making environment faced by a firm, the management literature has adopted evolutionary biologist Stuart Kauffman's NK model Kauffman and Weinberger (1989).

The NK landscape is defined as follows. There are N binary decisions ($\{1, 0\}^N$). The fitness of a position in the landscape is the average of individual fitness contributions which are a function of the current decision location, $\langle \ell \rangle$, $\sum_{i=1}^N c_i(\langle \ell \rangle) * \frac{1}{N}$. K determines the interdependencies, for each individual decision, ℓ_i there are K separate decisions which influence C_i ; thus $C_i(\langle \ell \rangle) = C_i(\ell_i, \ell_{k_{i1}}, \dots, \ell_{k_{iK}})$ where ℓ_{k_i} are the neighbors such that $\ell_{k_{ij}} = \ell_{i+j}$ except where $i + j > N$ in which case $\ell_{k_{ij}} = \ell_{i+j-N}$. Each contribution's fitness is drawn randomly from the uniform distribution over $[0, 1]$.

AR Length

A key feature of landscapes is the autocorrelation lengths, the correlation is fitness scores between locations of varying distances. Weinberger (1990) first noticed the autocorrelative properties of NK landscapes. The autocorrelation of NK landscapes decreases exponentially, and as K increases autocorrelation also decreases. While we may expect these properties, they create a different kind of ruggedness. As K increases we may notice that the number of maxima increase, but the landscape starts to resemble a collection of random draws rather than any systematic system; one may go from a global maxima to minimum in just one step. This makes search more difficult and less strategic or algorithmic as one cannot systemically search a landscape without correlative patterns. As we can see in the figure below, as the number of maxima increases (which is increasing in K) the autocorrelation decreases.

[Figure 1 about here.]

Local Area Networks

Kauffman's NK model may also be characterized by local optima networks (LONs) (Herrmann et al. 2016). LONs are clusters of landscapes optima which are reachable from other optima. For an optima to be within reach of another optima it must be possible to engage in hill climbing search from one optimum to another after being displaced by a step size of 2. A network of which optima are reachable from others is constructed and Markov clustering is applied to generate LONs.

LONs can characterize the relationship of the global optimum to other local optima. In the NK model, as K increases and as the number of optima increase we find that proportion of local optima that are in the same cluster as the global optima is relatively low and that the number of clusters increases. This suggests that many points in the fitness landscape are not able to reach the global optimum through moderately large step sizes. Indeed this suggests that larger step sizes and higher variance search strategies may be more helpful in the standard NK model as one would have to hope to first get into the global optimum's cluster before being able to reach it.

[Figure 2 about here.]

Effect Sizes

Another way of explaining a landscape is through regressions. We may Regress on each of the N variables, and potentially their interactions, or polynomials. If R squared is relatively high then that tells us that relatively good options can be found from a few variable combinations. Additionally if we find that the difference in prediction between polynomial degrees is small then that tells us that the interactions do not change the results quite so dramatically. The figure below illustrates that for Kauffman's NK model we see predictive power decrease with K , but also that there are large gaps between polynomial degrees suggesting that a few interactions do greatly matter for explaining variation.

R^2 Analysis

An important way to understand a complex landscape is to note the proportion of variance that may be explained by each component and the interactions of components. For each bitwise location $x_1, x_2, \dots, x_n \in X$, we can estimate the proportion of variance explained by regressing landscape fitness on each bitwise variable location with an ensuing β_X predictors corresponding to $Y = \beta_o + \beta_{X^1} \vec{x}_1$. We can also include a vector for all binary interaction terms $\vec{x}_2$, that is the vector corresponding to each component of: $X^2 = X \times X = \{(x_1, x_1), (x_1, x_2), \dots, (x_{n-1}, x_n), (x_n, x_n)\}$ or the polynomial of degree 2. Importantly, as more of the variance can be explained by polynomials of lower degree, the landscape becomes easier to search. While 100% of the variance is guaranteed to be explained by a linear model of the form $Y = \beta_{X^{N-1}} \vec{x}_{N-1}$, the model consisting of $2^N - 1$ predictors corresponding to all the bitwise combinations of variables corresponding to the set X^{N-1} ¹.

[Figure 3 about here.]

In thinking about real-world decision making landscape that organizations and managers face, one may also expect main effects to be somewhat dominant. For instance, consider a prototypical company offering a platform for consuming cultural goods (movies, tv, games, food delivery, etc...) and facing multiple decisions regarding pricing decisions, customer support quality, number of goods offered, recommendation algorithm choice, etc. We may naturally expect there to be an interaction between the monthly subscription cost for a service, the quality of customer support and number of goods - though there may be interaction effects (customers who pay a lot may expect better customer support), price alone may dominate the interaction of price and customer support and the interaction of price and customer support quality may dominate the three way interaction of price, customer support and number of goods offered. However we may not expect some 3 or 4 way interaction to be of huge concern - the relationship of the interaction of the recommendation algorithm, price, goods offered, and service quality should not dominate the effect of price alone or the algorithm alone.

¹For Kauffman NK landscapes, the number of variables need to explain all the variance corresponds directly with K; for $K = k$, X^{k+1} predictors suffices to explain 100% of the variance.

When searching too, we may then expect that despite the existence of ruggedness in local peaks, that individuals may engage in differing search strategies when main effects exert less dominance. A landscape where main effects are very weak is one in which navigation unpredictable and we may then expect search strategies on this landscape to also be erratic. As such, we may expect that when ruggedness is held constant but main effects are more dominant in one landscape than another that search is more likely to be local. That is individuals do not alter their choices to such a wild degree because they may notice that changing one or two variables suffices to gather information or alter performance in an understandable fashion.

The Dirichlet Dot Product model

In light of the above macroscopic characteristics of the *NK* model, we propose the Dirichlet Dot Product (DDP) landscape to test our core hypothesis. A formal definition is given below, but the core intuition behind the landscape is that the fitness contributions of interdependencies are most often subordinate to the main contribution of the state of each characteristic. Adding a social media team within a firm's marketing department will have knock-on effects because of interdependencies, but the primary source of change in organizational performance will come from the presence of the team, not just the interdependencies with other elements. The point of the *NK* model is that interdependencies matter, and in the DDP model this remains true. However, our regression analysis above shows that the interaction effects are significantly larger than the main effects in the *NK* model. We believe that such a state of affairs is not common in the environments humans deal with. Dominance of the interaction effects is akin to saying that having better individual team members is far less important than the particular combination of team members. This would mean that a team randomly chosen from a recreation league will have an average performance equal to one chosen from the pool of active NBA players. Instead, we take the view that the independent quality of the team members needs to remain the primary contributor to performance outcomes, even if interdependencies also impact performance.

We show in the following that it is possible to maintain such an ordering of effect sizes while also varying the importance of interaction effect sizes and generating ruggedness. Furthermore,

any landscape generation process that maintains the importance of the main effects is likely to maintain some of the other macroscopic characteristics of single-peaked landscapes; because the state of a characteristic has a substantial main effect, the fitnesses between adjacent points will have higher correlations than in high K (rugged) NK landscapes. This will lead to the long autocorrelation lengths that are typical of NK landscapes for low values of K instead of the precipitous drop-off in autocorrelation observed in high K landscapes (see Figure 5). Likely consequences of this change are that i) the local optima networks (LON) will have fewer clusters, and ii) the cluster containing the global optima will constitute a significant fraction of the network. Both parts of this claim follow from the observation that higher correlations between adjacent points means it is less likely that points on a path between any two local optima exhibit a sequence of decreasing fitnesses (i.e., “escape paths” are more likely).

To create such an ordering of interdependencies, our model connects the calculation of fitness scores directly to the state of the characteristics (that is, the location in the combinatorial space) and then adds a stochastic component. By linking the fitness to the location, we ensure that adjacent points are more likely to have similar fitnesses. The stochastic component comes via a draw from a Dirichlet distribution, which we now describe before defining the model.

The Dirichlet distribution is the multivariate Beta distribution, which is defined by N , the number of dimensions or components, and α , a vector of length N of positive reals $\{\alpha_1, \alpha_2, \dots, \alpha_N\}$. A single draw from the distribution yields a vector $\mathbf{X} = (X_1, X_2, \dots, X_n)$ such that $\sum(X_i) = 1$. Each α_i in this α vector of *concentration* parameters determines the mean of the values along the corresponding dimension of draws from the distribution. The mean is $\frac{\alpha_i}{\sum_j \alpha_j}$. Accordingly, when α_i is the same for all i , the expected value of a draw from the Dirichlet is $1/N$ for all components. If the α_i are not all the same, the dimensions corresponding to higher α ’s have greater expected values. However, for our present proposal, variance is the more relevant moment of the distribution because it allows for the tuning of ruggedness in the model. $Var[X] = \frac{\tilde{\alpha}_i(1-\tilde{\alpha}_i)}{1+\alpha_0}$ where $\alpha_0 = \sum \alpha_i$ and $\tilde{\alpha}_i = \frac{\alpha_i}{\alpha_0}$. Thus, as α_0 —the sum of the concentration parameters—increases, the variance of each dimension decreases. We justify our choice of this distribution and its implications below, and now formally define the Dirichlet Dot Product landscape.

Let $\vec{E}_N$ be a Euclidean vector space on $\mathbb{R}^N$ and let $Dir(\alpha)$ be a Dirichlet distribution of dimension N parameterized by α . N is the same positive integer for both the vector space and the Dirichlet distribution. It corresponds to the dimensionality of the search problem, as in the NK model.

Let location $\langle \ell \rangle \in \vec{E}_N$ and $X = [X_1, X_2, \dots, X_N]$ be a vector of random variables drawn from $Dir(\alpha)$.²

The fitness of location $\langle \ell \rangle$ is:

$$F(\langle \ell \rangle) = \langle \ell \rangle \bullet X \quad (1)$$

where $\bullet$ denotes the dot product. In expanded notation, $F(\langle \ell \rangle) = \ell_1 \cdot X_1 + \ell_2 \cdot X_2 + \dots + \ell_N \cdot X_N$.

N and α are the only model parameters. For example, if $N = 3$ and $\alpha = (1, 1, 1)$, a draw from $Dir(\alpha)$ could be $(0.569, 0.102, 0.329)$. Using that draw with location $\langle 1, 0, 1 \rangle$, $F(\langle 1, 0, 1 \rangle) = 1 \cdot 0.569 + 0 \cdot 0.102 + 1 \cdot 0.329 = 0.898$. The model uses a new draw from $Dir(\alpha)$ for each location, so $F(\langle 1, 0, 0 \rangle)$ is not 0.569, but—if the new draw is $(0.435, 0.41, 0.155)$ —0.435. In this case, the change in fitness by moving from $\langle 1, 0, 1 \rangle$ to $\langle 1, 0, 0 \rangle$ is $0.898 - 0.435 = 0.463$. Of the overall decline, 0.329 comes from the change of the third component from 1 to 0 (i.e., the main effect), and the remaining 0.106 is accountable to the interactions across the components.

We tune the ruggedness of the landscape using the values of α . When using the same value for all dimensions, the mean draw for each component is the same and therefore the expected fitness contribution of each component remains the same as α changes. What produces ruggedness is the change in the corresponding variance. For $N = 3$ and $\alpha = (10, 10, 10)$, the expected variance is $2/279$. Because of this small variance, the contributions of the interactions will be small in expectation and neighboring points are likely to have similar fitnesses, implying local optima will be uncommon. When $\alpha = (1, 1, 1)$ the expected variance for each draw

²Because it is defined for $\vec{E}_N$, the DDP landscape can encompass continuous space, unlike the NK landscape, which is constrained to the N dimensional hypercube, i.e., the combinations of $0, 1^N$. Being defined over continuous space is benefit of the DDP model, but for the present analyses we focus only on the hypercube to make the necessary comparisons to the NK model.

from $Dir(\alpha)$ is $2/36$, which creates more variation in the fitnesses of neighboring points and is associated with many local optima. Note that the Dirichlet distribution is defined for any positive real number α , and as $\alpha \rightarrow 0$, the variance further increases. However, at these values, the mass of the distribution shifts to the corners of the support and draws result in a single dimension having nearly all the weight. This leads to a bimodal distribution, with fitnesses close to either 0 or 1. While potentially applicable for some scenarios, we focus on values of α for which the resulting fitness distributions are uni-modal around a value of 0.5. For $N = 10$, this is $\alpha \gtrapprox 0.5$

The Dirichlet distribution is a convenient and natural way to introduce dependencies between components. Because the values of $Dir(\alpha)$ sum to one, the larger an individual element is, the smaller the others have to be in summation. It is useful to think of a draw from the Dirichlet distribution as describing the probability of one of N different mutually exclusive events occurring, such as in the rolling of a (weighted) dice. Because of mutual exclusivity, as one event becomes more probable, the others become less likely, meaning there is conditional dependence between the events. This dependency across the components is the key feature of the distribution. By directly tying the values of the dependencies to the source of stochasticity, the relationship is more transparent than the summations of uniform random draws on theoretically inert substrings the NK model uses. Other aspects of the distribution make it a natural choice. By using asymmetric concentration parameters (e.g., $\alpha = (5, 1, 3)$) it is possible to tune the means and relative variance of each component's fitness contribution. We do not make use of this feature here, but it makes it possible to make particular dimensions more important contributors to the fitness outcomes. Finally, the Dirichlet distribution is also the conjugate prior of the categorical distribution, which could be useful for modeling Bayesian learning of landscape features given the observation of discrete outcomes.

There are a few other features of this model that should be noted. Most importantly, by linking the fitness to the location in the vector space, the model requires a new interpretation of the locations in the vector space. Locations in the NK landscape are typically labeled with bits ($[0,1]$), but the model does not suggest that the distances between states are meaningful. The labels could just as well be A and B or T . In the DDP model, the states of a dimension

are meaningful, not just a label. While state pairings of 0 and 1, -1 and 1, or 1 and 10 are substantively equivalent in the NK model, each pair will alter the features of the landscape in the DDP model by controlling i) the contributions of different states of the components and ii) the relative contributions of different components. This opens up the parameter space of the landscape for novel analyses, but also requires empirical or theoretical interpretations of the state labels.

Relatedly, once the minimum and maximum state values for each dimension are defined, the minimum and maximum fitness values of the landscape are known. For example, if the two possible states of an individual component are 0 and 1, the minimum fitness for $N = 3$ is $F(\langle 0, 0, 0 \rangle) = 0$ and the maximum is $F(\langle 1, 1, 1 \rangle) = 1$. Crucially, these two values are the *maximally distance* locations within the space. This feature was a chief inspiration for the model; when there are no interdependencies in the NK model (i.e., $K = 0$), the landscape is single-peaked and the fitness minimum and maximum are the maximum Hamming distance apart³. The DDP model always maintains this macroscopic fitness gradient, while adding local optima between the two extremes. The NK model does not do this and, in fact, the maximum and minimum fitnesses can be neighboring points for low values of K .⁴

Another key feature is how exactly the Dirichlet draws become the source of fitness interdependencies. The contribution to fitness of a given component depends on the states of all components because each location is associated with a unique draw from the distribution. To see this clearly, let $F(\ell_j)$ be the fitness of a location ℓ_j and let X_{i,ℓ_j} be the value of the Dirichlet draw X_{i,ℓ_j} for the i^{th} dimension of location ℓ_j . Now consider the location $\ell_1 = \langle 1, 0, 0, 0, 0, 0, 0, 0, 0, 0 \rangle$. The fitness of the location, F_{ℓ_1} , is simply the value of the draw for the first dimension, X_{1,ℓ_1} , because all other dimensions are multiplied by zero before being added. Next, consider $\ell_2 = \langle 1, 1, 0, 0, 0, 0, 0, 0, 0, 0 \rangle$. The second dimension of the draw, X_{2,ℓ_2} is now added to the overall fitness to get the $F(\ell_2)$. But, crucially, $F(\ell_2) \neq F(\ell_1) + X_{2,\ell_2}$

³It is natural to think of a single peaked landscape as having a literal peak of a global maximum surrounded on both sides by points with lower fitness values, but if we could project the fitness points onto a single location dimension, the correct image of a $K = 0$ landscape is instead a line with a monotonic slope. Because the fitnesses of the dimensions are independent, the fitness contributions of each come from two draws from a uniform distribution. To find the global maximum, one needs to simply identify which state is associated with the larger of the two values for each dimension. Once identified, the global fitness is necessarily the location that inverts each state to their minimum value.

⁴As explained below, in the experiment we transform the landscape to hide this feature from participants.

because X_{ℓ_2} is a different draw than X_{ℓ_1} and the contribution of first dimension is changed (i.e., $X_{1,\ell_1} \neq X_{1,\ell_2}$).

The expected increase in fitness through the change of the second component from 0 to 1 is greater than the expected change in the contribution of the first component. Consequently, the main effect of changing the state of the second component to 1 is positive. However, due to the stochasticity supplied by the unique draws from the Dirichlet, there is a difference between the total fitness of the location $F(\ell_2)$ and the fitness of $\ell_1 + X_{2,\ell_2}$. This difference, $X_{1,\ell_1} - X_{1,\ell_2}$, is the magnitude of the interaction between components 1 and 2 in this case. It can be either positive or negative, corresponding to positive or negative complementarity between the components. In the appendix, we discuss how to extend this logic to understand the contributions when many components change.

Finally, in order for a landscape to be complex, there need to be negative complementarities. If there are none, the landscape is necessarily single-peaked. The presence of negative complementarities increases as α decreases in the DDP model, but the magnitudes are more constrained than in the NK model. In the NK model, the sizes of the complementarities increase in K and, as shown above, therefore have larger magnitudes than the main effects. This yields ruggedness, but also wide swings in fitness and a disintegration of broad trends in fitness. The DDP model avoids this fate in part because of the smaller magnitudes of the complementarities.

Having defined the model, we now turn to analysis of features of the landscapes it produces. We are arguing that several macroscopic features of the high- K NK landscapes other than ruggedness explain the ineffectiveness of search in those landscapes. A first step in demonstrating this is to confirm that the DDP landscape can tune ruggedness without altering the other features. The degree of ruggedness produced by a given α value depends on the dimensionality of the problem space, or N , and here we focus on $N = 10$. Figure 4 shows that the DDP landscape can create approximately the same degree of ruggedness as the NK landscape. The regions of interest for the domains of α and K are not the same, so the respective domains are plotted on the two different Y-axes. The number of local optima is plotted on the X-axis. The plot reveals that α can be tuned to allow the DDP models to have a similar number of local optima as the NK model does. The relationship between α and the count of local optima is

not approximately linear as it is for K , but the DDP model covers the domain of the NK model. Crucially, we find that $\alpha > 15$ produces single-peaked landscapes and that $\alpha \approx .1$ produces a similar number of local optima as is expected for $K = 9$ in the NK model.⁵ In the experiments, we use landscapes with identical counts of local optima for direct comparison.

[Figure 4 about here.]

Because the two models can produce landscapes with the same degree of ruggedness, we turn to the macroscopic features discussed above. While the ability to completely control the other features would be ideal, they have substantive connections and the variation in macroscopic features between the two models is ultimately a matter of degree.

With this in mind, Figure 5 shows that autocorrelations lengths are longer for the DDP model than for the NK model. When $N = 10$, the line for $\alpha = 15$ is more or less identical to the line for $K = 0$; the autocorrelation coefficient for a distance of 4 is approximately 0.5 in both. In fact, the DDP model with $\alpha > 15$ produces a landscape that is more or less identical to a single-peaked NK landscape. However, while increasing K quickly drives autocorrelation to zero, decreasing α to produce the same number of local optima as $K = 9$ does not. $\alpha = 3$ has an autocorrelation coefficient of 0.6 at a distance of 1 and 0.3 at a distance of 4. This is approximately twice the value for $K = 4$, but those landscapes have the same expected number of local optima. This implies that a greater amount of information about neighboring fitness values is contained in the fitness of a given point in the DDP. Contrast this with the coefficient of zero at a distance of 1 for $K = 9$. Whereas the fitness value of a location in highly rugged NK landscapes contains no information about the fitnesses of neighboring locations, the highly rugged DDP landscapes are more like landscapes $K = 6$ with respect to autocorrelation lengths. This is consistent with the existence of a meaningful maintenance of the steady gradient from minimum to maximum found in single-peaked landscapes.

[Figure 5 about here.]

⁵The lowest values of α are not included because below those levels, the mass of the distribution is always at the corners of the simplex. This means the draw is a “one-hot” vector and only one dimension contributes to the fitness. Although such “holey” distributions may exist in genetic evolution (Dochtermann et al., 2023), they are outside the scope of this paper.

By maintaining a broader gradient, the DDP model also succeeds in not fracturing the landscape up into many funnels within the local optima networks, as defined by Herrmann et al. (2016). Our own funnel analysis shows that when $\alpha = 1$, the DDP model produces landscapes with an average of 1.66 funnels (i.e., LON clusters) and that the funnel containing the global maximum contains more than 97% of all locations. By contrast, our analysis of $N = 10$, $K = 9$ landscapes show them to have 8.25 funnels on average and that the funnel containing the global optima includes an average of 33% of all points. In terms of funnels, the $\alpha = 1$ landscapes are comparable to $K = 1$ NK landscapes, further suggesting that rugged DDP landscapes will be associated with better search outcomes.

Underlying these differences are substantially different magnitudes of interaction effects. Figure 6 replicates the R^2 analysis found in figure ?? . For the most rugged landscapes we analyze, $\alpha = 1$, the state of the component itself, that is, the main effect, accounts for approximately 55% of the variation in fitness. As discussed in the previous section, a consequence of using the Dirichlet distribution as the source of variation is that the model creates interdependencies across all components and then varies the strength of the interdependencies. This is why the second- and third-order polynomials do not explain meaningfully more variation. The use of asymmetrical concentration parameters or different subspaces of $\vec{E}_N$ could increase the explanatory power of those terms if so desired, but the primary point is that the main effect of the state of the component is a meaningful source of variation in fitness scores. That is clearly the case here for all degrees of ruggedness.

[Figure 6 about here.]

The above analyses show that the DDP model can tune ruggedness while preserving important macroscopic features of single-peaked landscapes. This makes it a great candidate landscape for isolating the ruggedness from the other features and exploring how it impacts search behavior and outcomes, which we now do.

The Animal Art Game

The previous section demonstrates that the DDP model creates rugged landscapes with macroscopic features closer to single-peaked NK landscapes than comparably rugged NK landscapes.

This variation allows us to better understand the role of ruggedness itself in limiting the effectiveness of search. A natural way to explore these differences would be to use agent-based models, as is typical in *NK* literature. However, there are numerous plausible human-like search heuristics as well as the possibility that real human search and learning behaviors are much more complex than can be easily encoded. Rather than attempt to systematically revisit the alternatives found in the literature, we instead use a preregistered, incentivized human-subject experiment to capture a broad range of search and learning styles across both landscape.

Our experimental design is a slight modification of the “Alien” game introduced by Billinger et al. (2013). In this game, participants are incentivized to search a fitness landscape by toggling shapes on and off under the premise of creating art for evaluation by an alien. The payoff is a value that represents the aliens’ willingness to pay for that particular piece of art. The preregistered design and hypotheses are available at https://osf.io/9zqhy/?view_only=cc9fccd5271849689a97e70cc00757db and an image of the interface is available in the appendix.

We recruited 160 participants based in the United States from the Prolific online panel. After giving their consent and reading instructions, participants have 25 rounds to search for high-payoff combinations of shapes for $N = 10$. They can toggle on and off as many of the 10 shapes as they would like until they press submit (within the 30 second submission window). At that point, they are shown the payoff for that combination of shapes, and it is added to their point total, which is always displayed on the interface. Participants have readily available a full history of their submissions, visualized as the combinations of shapes and the points associated with that combination. Submissions are shown in a column with the most recent ones on top. This history allows them to more easily track the locations they have visited and was present in the interface in the original game of Billinger et al. (2013). Returning to those locations requires manually toggling the shapes in the main interface.

In all landscapes used, the fitness scores are scaled between 0 and 100 and participants are compensated \$0.001 for each point earned. Thus, for each submission at the global maximum of 100, they would receive \$0.10. There is a base compensation of \$2 for the task, and the instructions make it clear that receiving any compensation requires finishing the entire task. Completion of the whole task took about 15 minutes on average, and the average performance-

based compensation was approximately \$5.50.

We call our version the “Animal Art” game. Each participant searches three different landscapes and, to stress that each is a different landscape, we associate each with a different, unnamed species of animal. We had each participant search over multiple landscapes to make it possible to look at within subject differences, a topic this paper ultimately does not address. To avoid possible ordering effects in the data, each participant was randomly assigned one of 12 different sequences of landscapes. We generated candidate landscapes using different values of α and K ⁶ and created three different groups based on the number of local optima they contained. We use six “easy” landscapes that have a single local optimum, three generated using the *DDP* model and three using the *NK* model. Similarly, there were six with 33 local optima—the “medium” landscapes—and six with 84 local optima—the “hard” landscapes.

We use four basic sequences of landscapes. Because we expect, and pre-testing confirmed, that there would be no differences in outcomes among *DDP* and *NK easy* landscapes, those are treated as the same landscape type and are always the second in the sequence. Consequently, two basic sequences start a hard landscape of the respective types and end with a medium landscape of the opposite type (i.e., [*NK-hard*, *DDP-easy*, *DDP-medium*] and [*DDP-hard*, *NK-easy*, *NK-medium*]). The other two basic sequences start with medium landscapes and end with hard ones. By varying whether the game starts with a hard or medium landscape, we can assess whether there were improvements in search behavior across the rounds of the game and control for any ordering effects.

From those four basic sequences, we construct 12 by using different instances of each type to make sure that the effects are not the result of particularities of any one landscape. In our analyses, we look for those differences and find none, allowing us to combine the 12 sequences into the basic four sequence types.

⁶Because the generating process makes the point $\langle 1, 1, \dots, 1 \rangle$ the global maximum and $\langle 0, 0, \dots, 0 \rangle$ the global minimum, we apply a constant transformation to each location. Doing so preserves the fitness relationships between neighboring points, but disguises the location of the global maximum and minimum, lest that bias search behavior and outcomes.

Quantifying Effectiveness of Search

Following the literature, we analyze the effectiveness of the search through a combination of absolute and relative performance measures. Absolute measures of performance are certainly theoretically important and speak directly to the challenge of the problem. But without strong incentives to find the optimal solution, considering the relative ease with which participants find *better* solutions is another important way to gauge the difficulty of the problem presented by a particular landscape.

With these differences in mind, we look at several measures of average performance for all individual searches on the respective landscapes. The first measure is simply the fitness score of the current round. While straightforward, participants know the number of rounds remaining and this might induce satisficing behavior in which search is ended a few rounds prior to the completion of the game in order to maximize total payoffs. Alternatively, the search might continue and exhibit noisy outcomes of large jumps up or down in fitness right at the end. A rolling average would account for that, but could also hide important search behavior. In particular, after a failed search elsewhere, a participant might return to a known high-performance location right at the end of the game.

Rather than look at rolling averages, we follow Billinger et al. (2013) and look at the highest score found over the entire search process. This “highwater mark” speaks to the ease with which one can find high-scoring locations either through systematic local search or random “long jumps.” Successful search requires taking lower payouts, sometimes even extended time. This is true even in single-peak landscapes because participants do not know the scores associated with neighboring combinations until they visit them. The participant might remain very aware of the highwater mark, but, given the finite number of search periods, fail to exceed it or even return to that location or level. For this reason, this highwater mark measure is theoretically important for the analysis. It also makes our results comparable to the results of Billinger et al. (2013). We look at both “current” and “highwater” scores to make sense of search outcomes.

However, comparing these two measures between different types of landscape ignores the fact that the variance in the distribution of all fitness scores can be different between them. Larger

variances make it more likely that search starts from lower scores and reaches high scores. This is a key macroscopic feature of different landscapes, and it necessitates adjustments to make more direct comparisons between landscapes. Our first adjusted measure looks at the total improvements from the initial submission. To start the game, we place participants at a location with a score that is one standard deviation below the mean. We do not reveal the associated score, and many participants move before seeing it (i.e., they never “submit” from that location). In that regard, participants rarely start from the same position, but because of the correlation between location and fitness and an observed tendency to start with small moves, the initial starting score is likely to be lower for the wider variance landscapes. This makes the total improvement from that initial location and score an important measure of the success of the search process.

However, the larger variance also means it is easy to find higher scoring locations simply because there are more of them. Again, this speaks directly to differences in the structure of the landscapes, and, we argue, measures should not adjust this difference away. However, to highlight the difference, we also measure improvement in terms of change in the percentile rank of all scores. For example, the 97th percentile score in a low-variance landscape will be lower than the 97th percentile score of a high-variance the former but might be just as hard (if not harder) to locate than the former. For this reason, our fourth measure is the total change in percentiles of the scores.

With these measures of performance in mind, we rely on the Mann-Whitney U test to statistically assess differences. This test, also known as the Wilcoxon rank-sum test, looks at the probability that values drawn from one population are larger than values drawn from a second population. It accomplishes this by looking all pairs of values from the two distributions. When applied to interval-scale scenarios such as ours, this binary comparison of the U-test is more conservative than a T-test because the latter gives more weight to outliers. For example, a handful of extremely high scores for one population increases the probability that the T-test will reject the null that there are no differences even if most outcomes are the same across the populations. The U-test instead looks for evidence that more than 50% of the values of one population is greater than the second population and therefore extremely high scores do not

contribute to this possibility any more than marginal differences.

Hypotheses

Our core preregistered hypotheses are that search on the *hard* DDP landscapes is more successful than on *hard* NK landscapes, both in terms of final scores and performance increases. In an attempt to gauge how much better, we also hypothesize that search on *hard* DDP landscapes is *as successful* as *easy* NK landscapes. Note that the second set of hypotheses is not statistically testable. Instead, we test the alternative hypothesis that they are statistically different with the expectation that we will fail to reject the null hypothesis that they are the same. To be clear, this is not statistical evidence that search outcomes on the two types of landscape are the same, but failing to reject the null that they are the same provides (limited) insight into how much better search outcomes are on the DDP landscape.

H1a, b: The average scores (a) and average performance increases (b) for the last round of play are significantly greater for the *hard* DDP landscapes than the *hard* NK landscapes.

H2a, b: The average scores (a) and average performance increases (b) for the last round of play are not equal for the *hard* DDP landscapes and the *easy* NK landscapes.

Although not among our preregistered hypotheses, we also test for the same differences with the highwater scores to further highlight the differences in search behavior.

H3a, b: The average highwater scores (a) and average highwater performance increases (b) by the last round of play are significantly greater for the *hard* DDP landscapes than the *hard* NK landscapes.

H4a, b: The average highwater scores (a) and average highwater performance increases (b) by last round of play are not equal for the *hard* DDP landscapes and the *easy* NK landscapes.

WHAT HAPPENED TO THE MEDIUM LANDSCAPES?

Results

[Figure 7 about here.]

The results largely correspond to our theoretical intuitions and also begin to clarify some of the key features that determine when search is effective. Figure 7 plots the mean current score of all participants in the left panel and mean highwater score of all participants in the right panel. Crucially, H1a, that the final scores for hard *DDP* outcomes are significantly greater than for hard *NK* outcomes is supported ($p < 0.001$), demonstrating clearly that search on highly-rugged *DDP* landscapes is more effective than on highly-rugged *NK* landscapes. H3a, which makes the same comparison but using the highwater scores is also supported ($p < 0.001$).

However, while performance on the *DDP*-hard landscapes is approaching that of *NK*-easy landscapes, final scores for the two landscapes are in fact statistically different ($p < 0.01$); search outcomes on easy *NK* landscapes is easier than on hard *DDP* landscapes. This is true for the highwater scores too ($p < .02$). This rejection of the null means search on hard *DDP* landscapes is not as effective as we'd hoped. However, there is no statistical difference in highwater scores for hard and easy *DDP* landscapes ($p > .75$).

A potential reason performance on *DDP* landscapes was not as effective as we had hope for is that search on the hard *DDP*-hard landscape starts at a lower value than the other landscapes. This is visible in Figure 7. Figure 8 shows total improvements for both the current and highest scores across the treatments. This improvement measure makes clearer how much easier search on the hard *DDP* landscape is than the hard *NK* landscape. The average total increase in performance for hard *DDP* is greater than hard *NK* (H1b, $p < 0.001$) and is not statistically different from easy *NK* landscapes (H2b, $p > 0.9$). Not only are the highwater improvement scores for hard *DDP* better than hard *NK* (H3b: $p < 0.001$), they are better than easy *NK* (H4b: $p < 0.01$)!

[Figure 8 about here.]

The above evidence shows quite clearly that search on very rugged DDP landscapes is more successful than search on very rugged NK landscapes. By some measures, the search on the former is comparable to search on single-peaked *NK* landscapes, strong evidence that ruggedness alone need not be an impediment to effective search. However, the results offer hints of a more complex story; search on hard *DDP* landscapes generally outperforms search on easy *DDP* and *DDP-medium landscapes*. In the case of easy *DDP*, the final differences are fairly small in magnitude, albeit generally statistically significant. However, the magnitudes of the differences between medium *DDP* and hard *DDP* are often substantial. While search on medium *DDP* outperforms medium *NK* landscapes, there is a curvilinear relationship for the *DDP* landscapes in which outcomes get worse for the initial increase in ruggedness, but improve greatly for the second increase in ruggedness.

We now show that this surprising pattern can be explained by the higher order moments of the distributions of fitnesses generated by the two models. Whereas the variance of the fitness distribution contracts as K increases in the *NK* model, variance *increases* as α decreases (and creates more ruggedness) in the *DDP* model. This is a natural entailment of the use of the Dirichlet distribution; for low values of α , the density of the probability function moves toward the edges of the simplex so that the most likely outcome of a draw is that one component dominates the resulting vector (e.g., [.67, .17, .16] or [.19, .12, .71]). As this happens, the mass of the fitness scores tends to move away from the mean of .5. Figure 9 shows the CDFs of the easy and hard landscapes used in experiment and clearly shows how the curves for the three hard *DDP* landscapes approach linearity.

[Figure 9 about here.]

This broadening of the distribution of fitnesses could make search easier just by “enriching” the landscape. The more high fitness values there are, the easier they are to find. For this reason we also look at increase in performance in percentile scores. Figure 10 shows the plots for these measures in terms of both current and highwater scores. Viewed this way, it becomes clear that the richness of the landscape is a key determinant of search outcomes. The percentile-adjustment final scores for hard *DDP* and *NK* landscape are not statistically different ($p > .25$),

although the highwater scores are ($p < .001$). Interestingly, there is no statistical difference between easy and hard *NK* landscapes ($p < .001$).

[Figure 10 about here.]

As measured in percentiles, the rank ordering of search outcomes for the *DDP* landscapes corresponds to the intuition that search should get harder as ruggedness increases. Furthermore, much of the variation in highest score outcomes for the *NK* landscapes disappears. These combined facts suggest that indeed the richness of the landscape plays a larger role in determining outcomes than ruggedness does. Variance captures some of that richness but Figure ?? provide a better characterization of the shape of the upper tail of the distribution by plotting the normalized distance from the 75th, 90th and 99th percentiles to the global maximum.

[Figure 11 about here.]

For the *NK* model, the distances increase from $K = 0$ until about $K = 5$, after which they level off. For the *DDP* model, the distances for $\alpha = 100$ and $\alpha = 10$ are about the same, but there is a significant decrease in distances for $\alpha = 1$. This thickening of the tail of the distribution is likely a major reason that search on this landscape is so successful.

Discussion and Conclusion

The *DDP* model offers an important window into how performance outcomes are tied to features of landscapes. Because rugged *DDP* landscapes maintain more of the macroscopic structure of single-peaked landscapes than *NK* landscapes, search outcomes are better. This is true even when the fitness scores are adjusted to reflect their percentile rank. In this sense, the ruggedness of the *NK* model plays much less of a role in determining search outcomes than is assumed in the literature. However, it appears that part of reason for more successful search on the *DDP* landscapes is that the generating process increases the variance of the distribution of fitnesses. As a consequence, there are more, higher fitness locations available in the landscape. There are approximately equally more locations with lower fitnesses too and that means our participants started at locations with lower fitness values on average. By contrast, the ruggedness generating

process for the *NK* model *reduces* the variance of fitness scores, impoverishing the landscape and making it much harder for actors to find the higher value points.

Our final analysis attempted to correct for these changes to the landscapes by defining redefining performance in terms of the percentile rank of the fitness score. In terms of final scores, performance on rugged *DDP* landscapes is still superior to comparably rugged *NK* landscapes, but this is not the case for highwater scores. In fact, the average highest score found by participants on highly-rugged *NK* landscapes is nearly indistinguishable from those on single-peaked landscapes (0.795 vs. 0.808, respectively). This fact suggests that the problem of search behavior is less about finding high scores and more about not losing track of them. The tendency of participants to lose track of high value locations might be related to their perception of the gap between their current scores and the global maximum.

Because the results are substantively similar to those in Billinger et al. (2013), we provide analysis of the patterns of search behavior in the appendix, rather than here. Instead, we simply conclude that the literature on performance landscapes has not addressed sufficiently the realized structure of *NK* landscapes and we believe this is to the detriment of the enterprise. The intuitions for why ruggedness would negatively impact search outcomes are strong if one assumes actors are more or less myopic hillclimbers, but Simon's call for a cognitive science stressed the importance of understanding cognition in the context of the environments in which it is applied. Rugged *NK* landscapes are complex environments and it would be surprising if varying *K* to produce ruggedness did not affect other features of the landscape. Without a full accounting of those features, it is too soon to claim that ruggedness is an impediment to the search processes of human actors.

Drawing on existing work and some of our own, we have shown that a variety of macroscopic features of the *NK* landscape do change as *K* increases. There is a rapid breakdown in the correlation of fitness values between adjacent points and any broader gradient of change in those values. The resulting landscape is not a many-peaked mountain range as is often depicted, but rather an irregular bed of nails, with locations of high fitness towering above their immediate surroundings. Such a landscape represents an extremely hard search problem and also fails to isolate the effect of ruggedness on search outcomes, which in turn prevents us from

understanding the relationship between ruggedness and search effectiveness.

To begin to address this issue, we propose the Dirichlet Dot Product (DDP) landscape. By linking the fitness scores to the locations within the combinatorical space, the model is able to maintain a broad gradient in scores while simultaneously introducing ruggedness. We believe that the style of interdependencies the *DDP* landscape is built upon are appropriate for a wide range of problems but of primary importance at present is the fact that the model facilitates further exploration of the role of ruggedness. We approach that exploration with a set of pre-registered hypotheses and a human-subject experiment, which shows that ruggedness is not incompatible with effective search. Outcomes on the most rugged *DDP* landscapes look very similar to those on single-peaked *NK* landscapes. It appears that this outcome is driven in part by the richness of the rugged *DDP* landscapes; in tuning the *DDP* landscape to be more rugged, it increases the number of locations with high fitness values. The more high value location there are, the easier they are to find and the more effective search is.

This suggests that the macroscopic features of performance landscapes play a more important role in determining search effectiveness than ruggedness does. It is not to say that ruggedness does not present challenges to local search, but given that human search heuristics involve taking big leaps into the unknown and the structure of that unknown determines the likelihood of success of big leaps, the search and literature needs to better understand that structure before linking behavior to cognitive faculties or psychological propensities.

Appendix

Assessing complementarities

It is possible to isolate the actual complementarities between two dimensions, but the fact that the values X_i change for each location makes doing so complicated. When we expand the above reasoning introduced in the definition of the model to all dimensions, $X_{1,\ell_1} - X_{1,\ell_j}$ is the *net* complementarity of component 1 with all other components. Isolating individual complementarities within that net complementarity requires traversing the shortest path from the location with a single 1 on the focal component, call it ℓ_H (for home), to location ℓ_j . The

differences in fitnesses at ℓ_H and the locations within a distance of one are the respective first-order interaction terms. The difference between the locations one step away and those two steps away, minus the first-order interaction term is the value of the second-order interaction and so on. Because of the exponential growth of the number of paths, the higher-order interactions approach the mean value of a Dirichlet draw. This suggests that first- and second order-complementarities make a difference, the bulk of the dependencies are manifested via the background noise of higher-order interactions, something our R^2 analysis above confirms.

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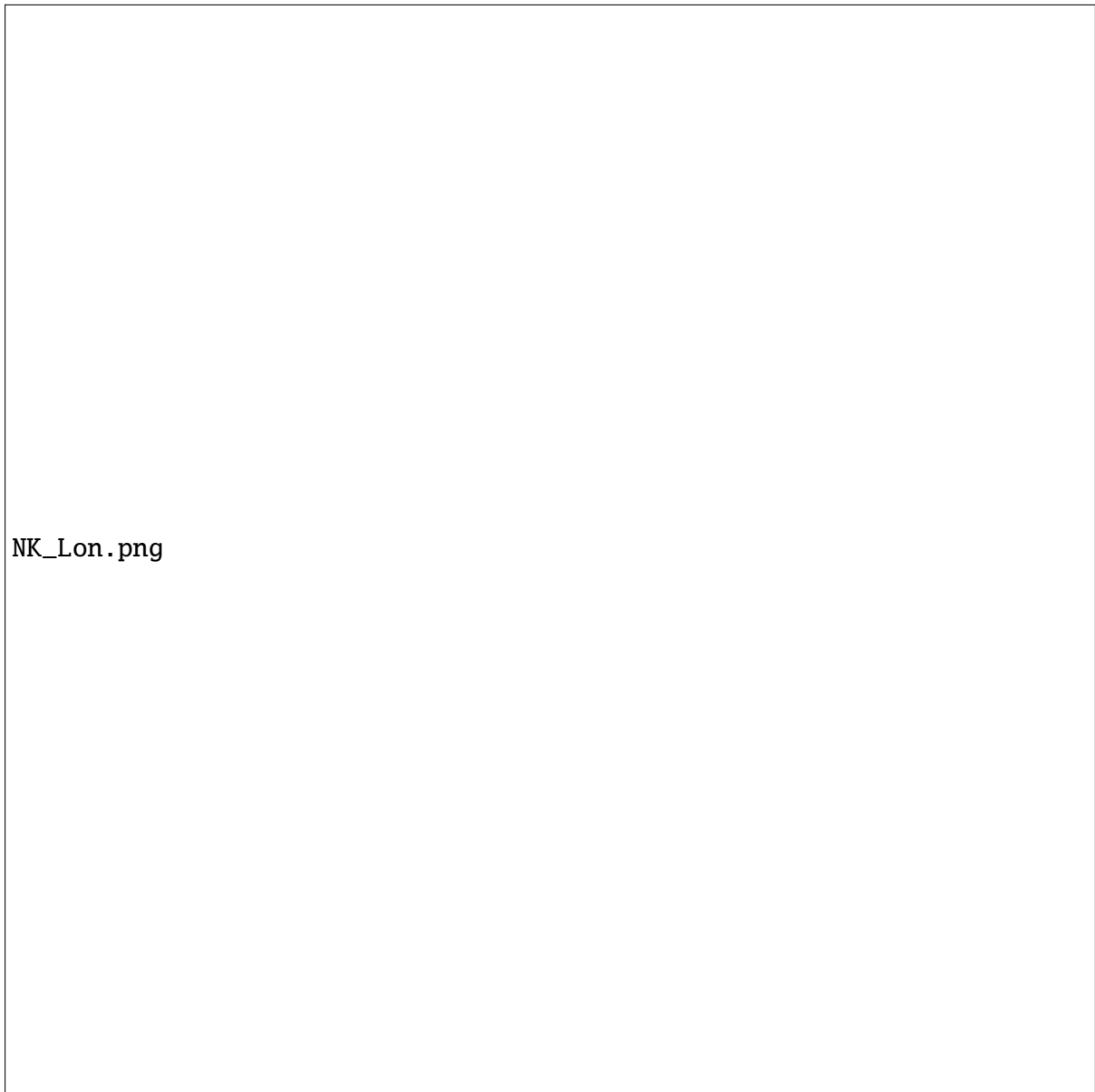
List of Figures

1	1 Step autocorrelation	32
2	NK Lon	33
3	R^2 and Ruggedness in the NK model	34
4	Ruggedness: The mean and variance for the count of local optima as a function of α and K . The values were found by creating 40 landscapes for each value of α and K	35
5	Autocorrelations by distance (steps) from the focal location for both the DDP and NK models. The coefficient of autocorrelation is calculated by averaging over a series of random walks over 40 landscapes for each value of α and K . The large dashed (black) line is for $\alpha = 0.1$ and the smaller dashed (blue) line is for $K = 9$, which have a similar number of local optima. The fact the former is above the latter indicates that neighboring points have more similar fitnesses in very rugged DDP landscapes than in comparably rugged NK landscapes.	36
6	DDP R^2 : R^2 analysis for the primary range of α . Each point represents an average R^2 value for 40 landscapes of the given value of α . The three R^2 's for each value of α correspond to regression models of 1st, 2nd, and 3rd degree (complete) polynomials, meaning all possible n^{th} degree interactions are included in the model.	37
7	Mean Performance: In the left panel, each line is the average of all scores for a given round for each landscape treatment. In the right panel, each line is the average of the each participant's highwater score (i.e., the highest score up to that round).	38
8	Increase in Performance: In the left panel, each line is the average of all scores for a given round for each landscape treatment. In the right panel, each line is the average of the each participant's highest score up to that point.	39
9	Cumulative Density Functions: The left panel contains the CDFs of the fitness distributions for all six easy landscapes used in the experiments. The right panel contains those for all six hard landscapes.	40
10	Increase in Percentile Performance: In the left panel, each line is the average of all scores for a given round for each landscape treatment. In the right panel, each line is the average of the each participant's highest score up to that point.	41
11	Percentile Distances: The distance from the given percentile to the global maximum for various values of K (panel <i>a</i>) and α (panel <i>b</i>)	42

Figure 1: 1 Step autocorrelation



Figure 2: NK Lon



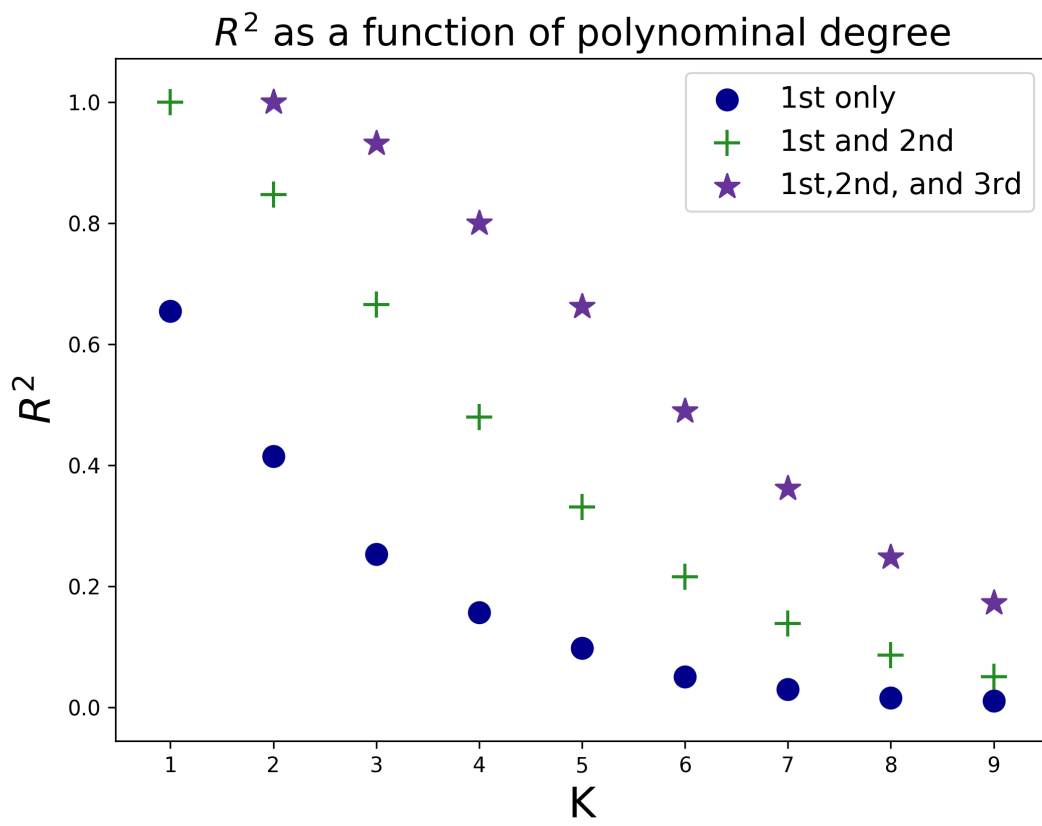


Figure 3: R^2 and Ruggedness in the NK model

Figure 4: Ruggedness: The mean and variance for the count of local optima as a function of α and K . The values were found by creating 40 landscapes for each value of α and K .

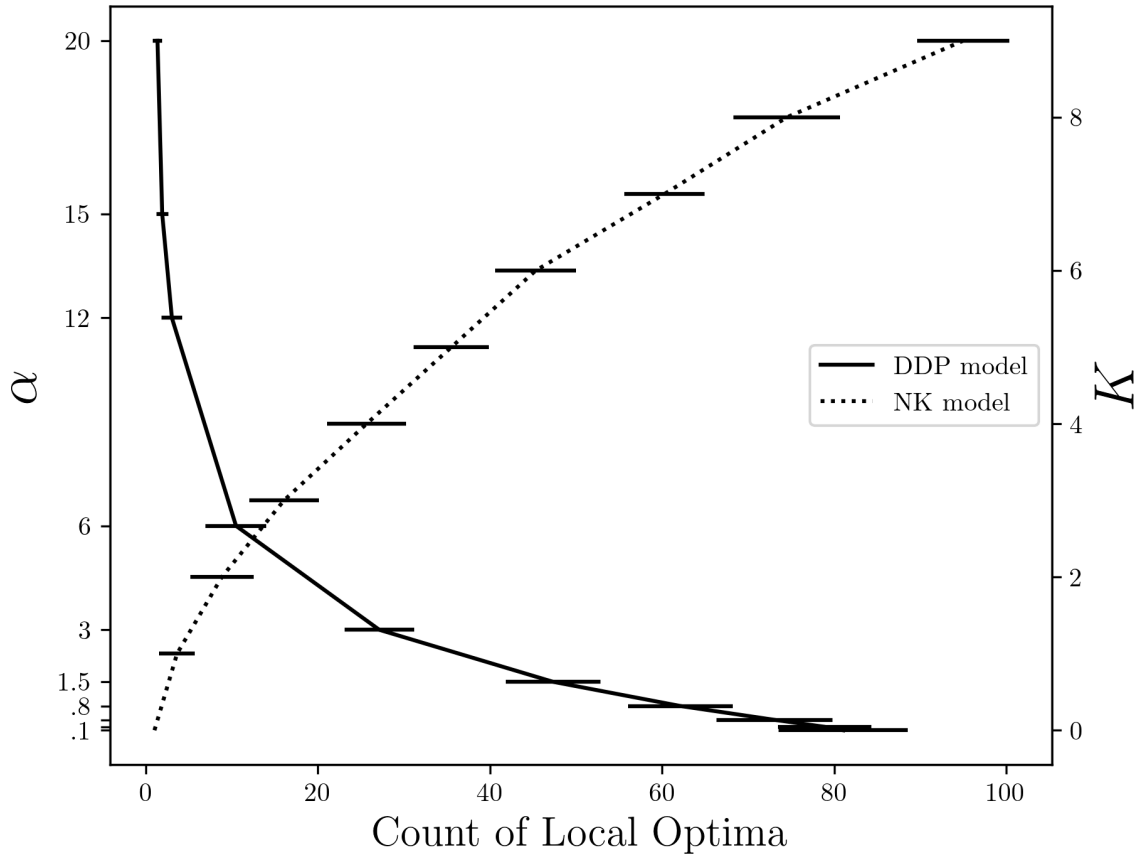


Figure 5: Autocorrelations by distance (steps) from the focal location for both the DDP and NK models. The coefficient of autocorrelation is calculated by averaging over a series of random walks over 40 landscapes for each value of α and K . The large dashed (black) line is for $\alpha = 0.1$ and the smaller dashed (blue) line is for $K = 9$, which have a similar number of local optima. The fact the former is above the latter indicates that neighboring points have more similar fitnesses in very rugged DDP landscapes than in comparably rugged NK landscapes.

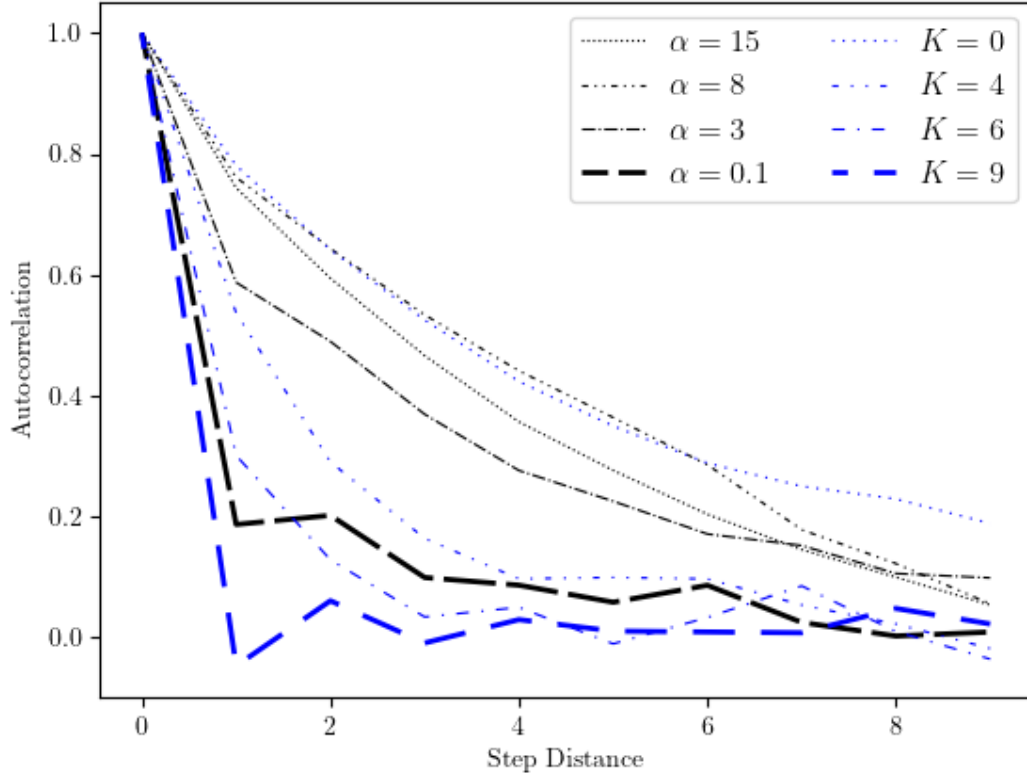


Figure 6: DDP R^2 : R^2 analysis for the primary range of α . Each point represents an average R^2 value for 40 landscapes of the given value of α . The three R^2 's for each value of α correspond to regression models of 1st, 2nd, and 3rd degree (complete) polynomials, meaning all possible n^{th} degree interactions are included in the model.

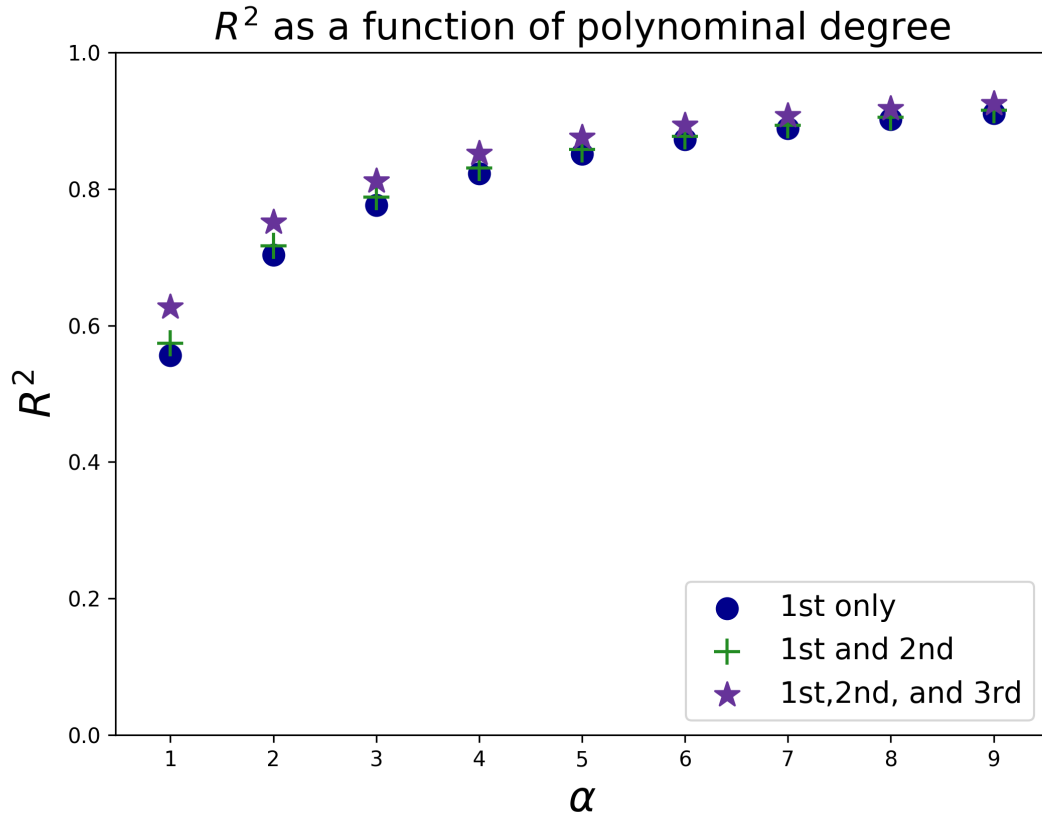


Figure 7: Mean Performance: In the left panel, each line is the average of all scores for a given round for each landscape treatment. In the right panel, each line is the average of the each participant's highwater score (i.e., the highest score up to that round).

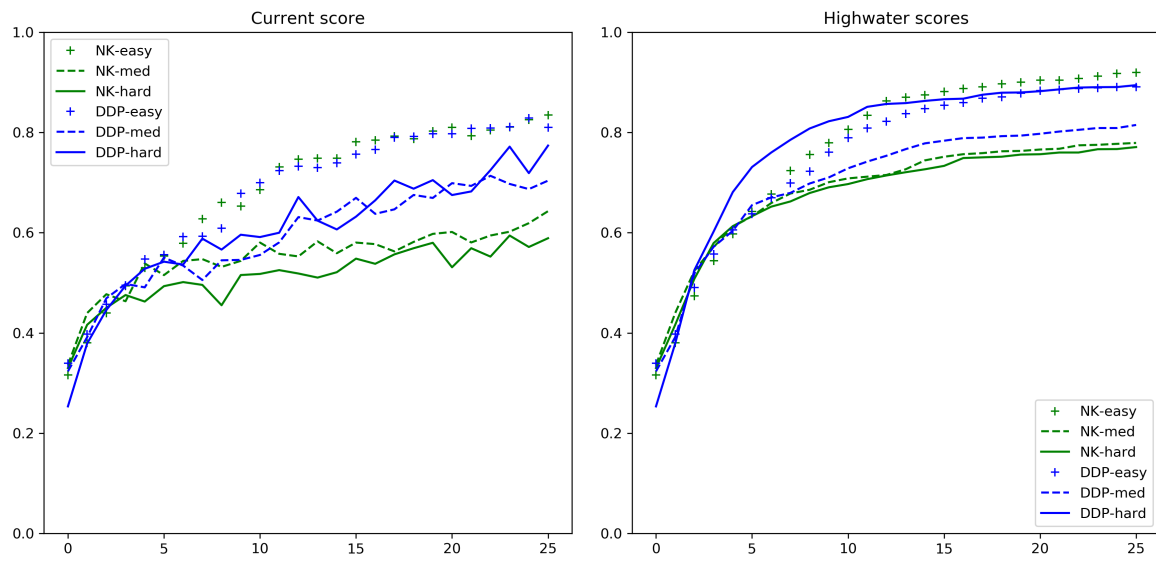


Figure 8: Increase in Performance: In the left panel, each line is the average of all scores for a given round for each landscape treatment. In the right panel, each line is the average of the each participant's highest score up to that point.

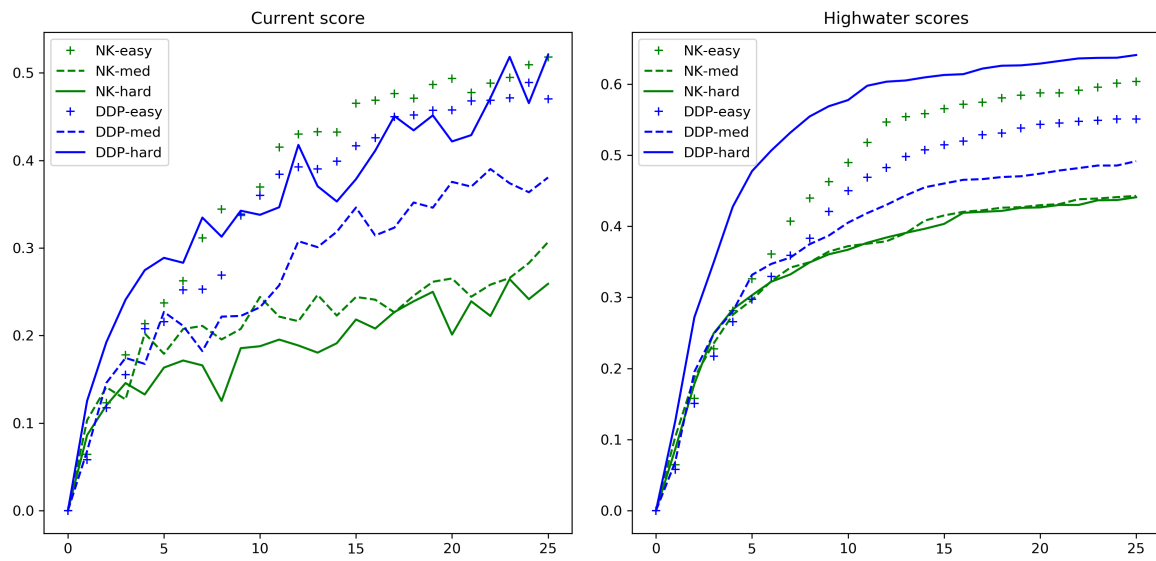


Figure 9: Cumulative Density Functions: The left panel contains the CDFs of the fitness distributions for all six easy landscapes used in the experiments. The right panel contains those for all six hard landscapes.

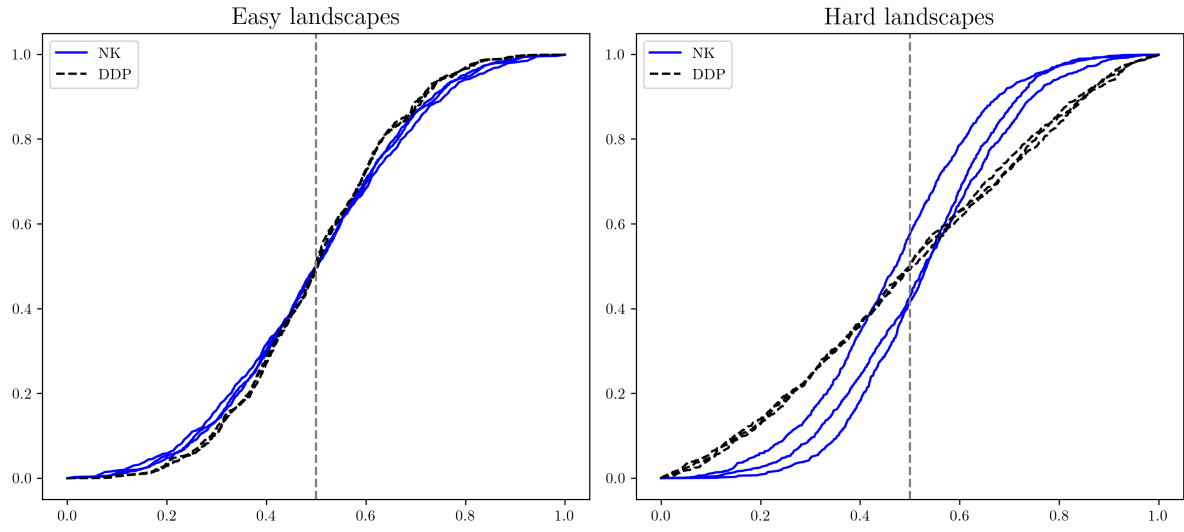
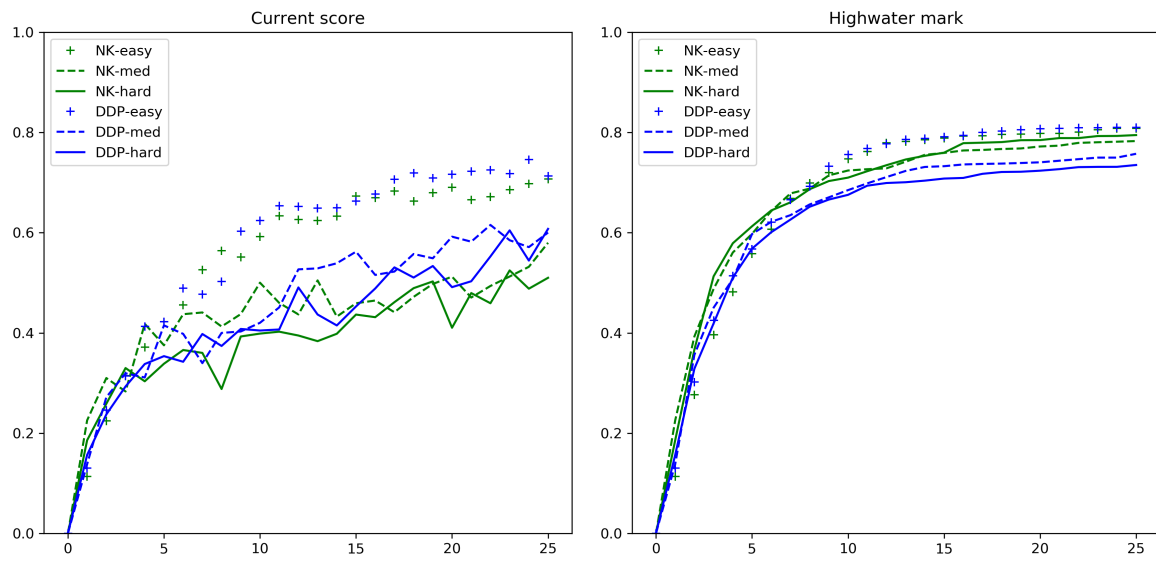
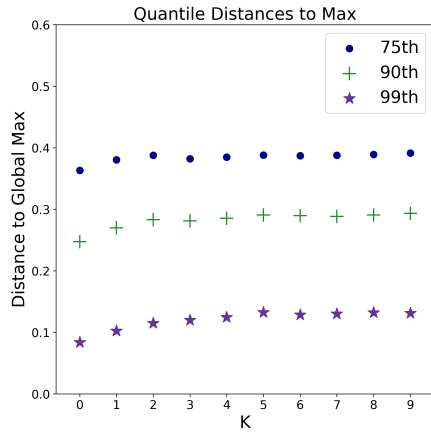
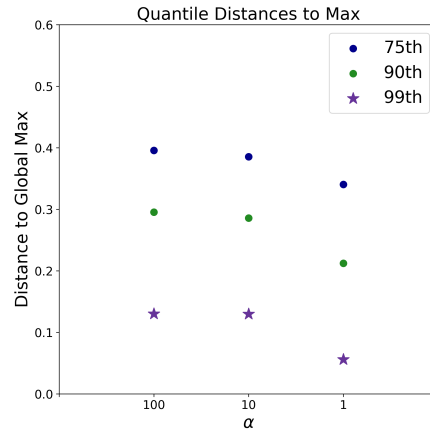


Figure 10: Increase in Percentile Performance: In the left panel, each line is the average of all scores for a given round for each landscape treatment. In the right panel, each line is the average of the each participant's highest score up to that point.





(a) hard NK percentile distances



(b) hard DDP percentile distances

Figure 11: Percentile Distances: The distance from the given percentile to the global maximum for various values of K (panel *a*) and α (panel *b*)